

Project Initialization and Planning Phase

Date	26 June 2026
Team ID	PNT2026TMID-EG01
Project Title	EduGenie - AI-Powered Educational Assistant
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

EduGenie is a lightweight AI-powered educational assistant designed to simplify and enhance the learning experience through Generative AI. It delivers intelligent educational support by helping students understand concepts, answer questions, generate quizzes, summarize learning materials, and receive personalized learning recommendations — all through a fast, interactive web interface built on FastAPI and powered by Google Gemini 1.5 Pro.

Project Overview	
Objective	Build an AI-powered educational assistant that helps students across all academic levels get instant, accurate answers, simplified explanations, personalized quizzes, text summaries, and structured learning roadmaps — making learning more interactive, accessible, and efficient.
Scope	The project covers the full-stack development of EduGenie, including backend API services (FastAPI), frontend web UI (HTML/CSS/JS), integration of Gemini 1.5 Pro and LaMini-Flan-T5 AI models, user authentication, learning history, and local deployment on resource-constrained devices such as Mac M1.
Problem Statement	
Description	Students and self-learners lack a single, unified platform that can answer academic questions, explain concepts at varying difficulty levels, generate topic-specific quizzes, summarize study material, and provide personalized learning paths — all in one lightweight, accessible tool.
Impact	Without EduGenie, students resort to scattered search engines, static course platforms, and generic chatbots that don't deliver educational context or structured learning guidance, leading to reduced comprehension, poor self-assessment, and lack of study direction.
Proposed Solution	
Approach	Build a FastAPI-based backend integrating Google Gemini 1.5 Pro for cloud AI and LaMini-Flan-T5 for local NLP tasks. Design a responsive HTML/CSS/JS frontend with distinct pages for each learning feature. Use SQLite with SQLAlchemy for data persistence and JWT for secure authentication.
Key Features	- AI-powered Question Answering (Gemini 1.5 Pro) - Simplified Concept Explanation at multiple difficulty levels - Topic-specific Quiz Generation for self-evaluation - Educational Text Summarization - Personalized Learning Roadmap with beginner-to-advanced progression - User Registration & Login with JWT authentication - Learning History tracking - Interactive Dashboard & Profile Management

Resource Requirements

Resource Type	Description	Specification / Allocation
Hardware		

Computing Resources	CPU / GPU specifications, number of cores	Apple M1 (8-core CPU, 8-core GPU) / Mac M1 local machine
Memory	RAM specifications	8 GB Unified Memory
Storage	Disk space for source code, SQLite DB, model weights, logs	256 GB SSD (local); GitHub for version control
Software		
Frameworks	Python web framework	FastAPI, Uvicorn
Libraries	AI / NLP libraries	google-generativeai, transformers, torch, SQLAlchemy, Jinja2, python-jose
Development Environment		
Data		
AI Models	Cloud & local AI models	Google Gemini 1.5 Pro (cloud API); LaMini-Flan-T5-248M (local HuggingFace)
User Data	Database engine and format	SQLite (edugenie.db) — stores users, quiz history, roadmap sessions